

Combinatorics – 5

Extremal combinatorics

14 сентября 2026 года

Theoretical minimum. Extremal combinatorics asks for the largest object without a forbidden configuration. The answers are few, short and worth knowing exactly; all of them are proved here.

Theorem 1 (Ramsey). For all s, t there is a least $R(s, t)$ such that every red-blue colouring of the edges of $K_{R(s, t)}$ contains a red K_s or a blue K_t ; moreover $R(s, t) \leq R(s-1, t) + R(s, t-1)$, whence $R(s, t) \leq \binom{s+t-2}{s-1}$ and $R(s, s) \leq 4^s$. *Proof* by induction, looking at the red and blue neighbourhoods of one vertex. The infinite version: every colouring of the pairs of an infinite set in finitely many colours has an infinite monochromatic subset. Lower bounds come from the probabilistic method (Probability – 4): $R(s, s) > 2^{s/2}$.

Theorem 2 (Turán). A graph on n vertices with no K_{r+1} has at most $(1 - \frac{1}{r})\frac{n^2}{2}$ edges, with equality only for the complete r -partite graph with parts as equal as possible. *Proof:* either by the “shift a vertex to its better-connected twin” argument, or by Cauchy–Schwarz applied to $\sum_v \deg v$. The case $r = 2$ is Mantel’s theorem: a triangle-free graph has at most $n^2/4$ edges.

Theorem 3 (Sperner and LYM). An antichain in the Boolean lattice $2^{[n]}$ has at most $\binom{n}{\lfloor n/2 \rfloor}$ members. *Proof:* count the maximal chains through each member of the antichain; distinct members give disjoint families of chains, whence the LYM inequality $\sum_{A \in \mathcal{F}} \binom{n}{|A|}^{-1} \leq 1$, and the bound follows.

Theorem 4 (Erdős–Ko–Rado). If $k \leq n/2$ and $\mathcal{F}$ is a family of k -subsets of $[n]$ any two of which meet, then $|\mathcal{F}| \leq \binom{n-1}{k-1}$. *Proof (Katona):* place $[n]$ on a circle; among the n arcs of k consecutive points at most k belong to $\mathcal{F}$, and averaging over all $n!$ circular orders gives the bound.

Theorem 5 (Dilworth and Mirsky). In a finite poset the minimum number of chains covering it equals the maximum size of an antichain (Dilworth),

and the minimum number of antichains covering it equals the longest chain (Mirsky, by grading with the height function). *Corollary* (Erdős–Szekeres): every sequence of more than $(r-1)(s-1)$ distinct reals contains an increasing subsequence of length r or a decreasing one of length s .

1. Prove the recursion $R(s, t) \leq R(s-1, t) + R(s, t-1)$ and deduce $R(s, t) \leq \binom{s+t-2}{s-1}$; compute $R(3, 3) = 6$ and prove the infinite Ramsey theorem. Then prove the lower bound $R(s, s) > 2^{s/2}$ by counting colourings, and compare it with the argument in Probability – 4.
2. Prove Turán’s theorem together with the description of the extremal graph, and deduce Mantel’s theorem. Then prove that a graph with n vertices and more than $\frac{1}{2}n^{3/2}$ edges contains a four-cycle, by double counting the paths of length two.
3. Prove the LYM inequality and Sperner’s theorem, and determine all antichains of maximum size. Then prove Dilworth’s theorem and deduce the Erdős–Szekeres theorem on monotone subsequences.
4. Prove the Erdős–Ko–Rado theorem by Katona’s circle argument, show that the bound is attained by the stars, and explain what happens when $k > n/2$.
5. Let $n \geq 2$ and let $\mathcal{A}$ be a family of subsets of $\{1, \dots, n\}$ such that $|A_1 \cup A_2 \cup A_3 \cup A_4| \leq n-2$ for any $A_1, A_2, A_3, A_4 \in \mathcal{A}$. Prove that $|\mathcal{A}| \leq 2^{n-2}$. (CIIM 2014; ФКН, весна 2026)